

NatureCheckpoint

IoT challenge for the city of Bilbao

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1. Description of the IoT system and its purpose

1.1. Main idea and target group

The idea of **NatureCheckpoint** is to create a network of small, weatherproof, solar-powered terminals placed in green spaces such as in urban parks, along the banks of the Ria in or along hiking trails to encourage outdoor activity in nature. Bilbao already publishes an Anillo Verde (GR 228) route dataset, which could support future planning of where to put the checkpoints [1]. Visitors collect a digital stamp by tapping their phone on each terminal. Stamps collected during a month unlock rewards, such as discounts at partner supermarkets and local shops, or free access to municipal activities. Exploring nature becomes a playful challenge that keeps people motivated to go outside, keep moving and spend more time in green spaces. The IoT system targets mainly students, residents, families and visitors of Bilbao and gives the city and local businesses a way to promote green spaces. Each terminal-device placed outdoor provides the following features:

- **Checkpoint:** Visitors tap an NFC-enabled phone on a marked terminal. The phone opens a lightweight mobile web app and requests a digital stamp for that station. Qualifying stamps collected over a calendar month unlock a reward, such as a discount from a participating local shop or access to a municipal activity, subject to real partner agreements. The app is used for seconds, not throughout the walk..
- **Dashboard:** Measures real-time environmental metrics (temperature, humidity, UV, air quality, noise, light, and pressure) to provide instant visual feedback via an integrated LED display.

The problem to test is not that Bilbao residents demonstrably avoid parks. Rather, our hypothesis is that some potential visitors face barriers such as a lack of motivation, established indoor routines and uncertainty about conditions. Urban green spaces provide opportunities for exercise, relaxation and social contact, but access by itself does not guarantee use [2], [3]. A 2024 review of gamified health applications reports modest improvements in physical activity rather than transformative effects, so our effect on walking habits must be evaluated rather than assumed [4].

1.2. Value created

- **Residents and students:** A low-pressure reason to take recurring outdoor walks, discover unfamiliar routes, obtain local environmental information and receive modest rewards.
- **Local businesses:** Optional sponsored rewards that encourage visits to nearby shops. Redemptions can be measured without revealing individual route histories to partners.
- **Bilbao City Council and park operators:** Measurements of local environmental conditions and participating checkpoint interactions, with aggregated statistics for comparing stations and promoting less-visited routes.
- **University and community organizations:** A route-based tool for welcoming newcomers, organizing outdoor events and testing ways to motivate healthier routines.

1.3. Semester MVP versus long-term solution

The **minimum viable product** will demonstrate one complete checkpoint and optionally a second station identifier. A Raspberry Pi 5 reads a practical subset of Grove sensors, serves or

communicates with a Python HTTP backend, drives an LED display and stores environmental data. Visitors use an NFC tag to open a mobile-friendly web page, collect stamps in a pseudonymous account and see progress toward a **simulated** monthly reward. A simple dashboard shows time-series measurements and valid check-ins. We will test both the sensor-to-display and NFC-to-server paths end to end.

Solar power, real sponsored vouchers, a city-wide deployment, outdoor enclosures certified for long-term use, LoRaWAN gateways, validated air-quality measurements and production-grade NFC anti-fraud are **scaling objectives**, not promises for the course prototype. This separation keeps the project achievable with the Raspberry Pi 5, Grove sensors and HTTP server used in the laboratory.

1.4. A typical real-life use case or scenario.

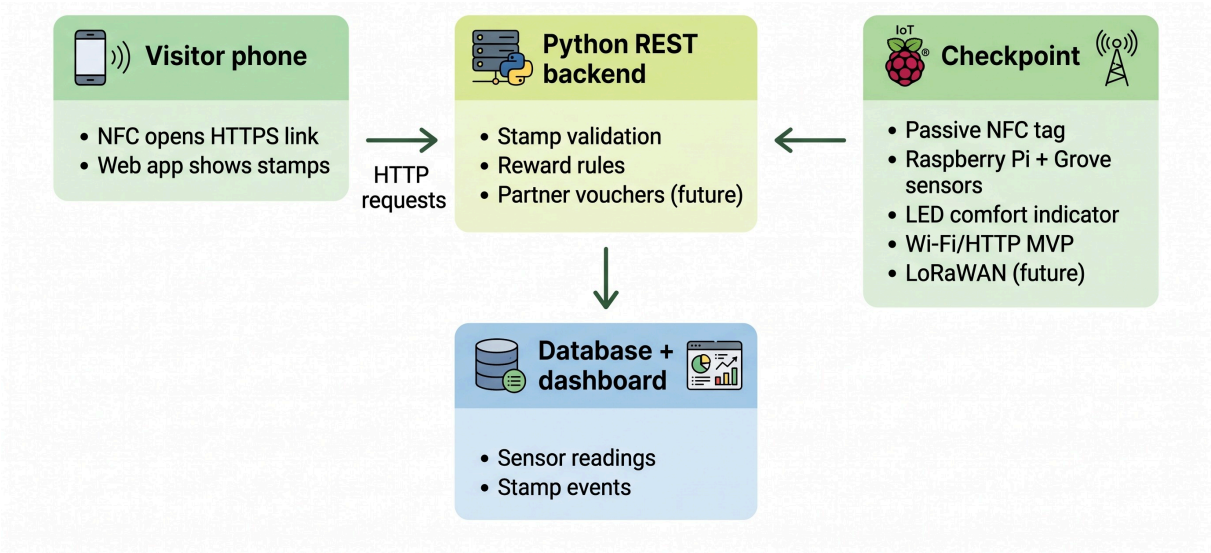
Scenario: Ana, a first year engineering student Ana has been in the library for hours. As she leaves, her phone shows a short, optional notification that the park on her way home is pleasant for the next two hours. She usually goes straight home, but she has five stamps this month and the checkpoint in that park is still missing from her route. Instead of immediately getting home and looking into the screen again she decides to take a break in the parque. So, she makes a ten-minute detour, reads the LED ring, taps her phone and gets "Stamp 6 of 10 this month". Then she puts her phone away, walks a slow loop under the trees, listens to the birds, and gets home about half an hour later than usual but more relaxed.

2. Advantages over market alternatives or existing systems

Alternative	Existing approach	Proposed difference
Actionbound	Mobile scavenger hunts with GPS, QR codes, missions and rewards [5].	Permanent IoT posts add measured environmental conditions and an on-site LED display.
Komoot / Strava	Digital route discovery or activity recording.	Stamp collecting does not require continuous GPS recording or sport-performance tracking.
Static trail signs	Information and nature interpretation at a fixed location.	Live microclimate readings and recurring monthly participation incentives.
Conventional loyalty cards	Visits or purchases unlock discounts.	Stamps are earned outdoors at green-space checkpoints rather than solely at shops.

NFC, web applications, sensors, loyalty programs and outdoor games already exist. NatureCheckpoint combines them around a different purpose: repeat visits to natural spaces with brief, location-specific feedback and support for local companies. With this project we are not inventing a new technology. The innovation is the combination and the context: using a physical, privacy-friendly check-in to link motivation, local conditions and local commerce around the single goal of getting people outdoors. The innovation is in the service that is provided and the value model, not in the hardware.

3. Solution architecture diagram



Stamp flow: the phone reads the NFC tag, which opens an HTTPS page, and submits a check-in. The server applies uniqueness and rate limits, stores the event and returns the new stamp count. The MVP uses a plain NDEF URL tag, which is enough to demonstrate the interaction but can be copied. The production design uses an NXP NTAG 424 DNA with Secure Unique NFC (SUN) messages, which authenticate every read [6], combined with server-side key, counter and fraud checks.

Sensor flow: Python reads the sensors at a fixed interval and discards missing or out-of-range values. The node updates its LED comfort indicator locally, without waiting for the cloud, and sends the readings to the HTTP service. The backend stores them with timestamp and station ID for the dashboard.

Key constraint: a passive tag cannot tell the node that a stamp happened, so the MVP confirms stamps on the phone only. A physical confirmation LED would need the server to notify the node (for example via MQTT) and would not work offline.

4. Sensors and actuators

Component	Priority	Purpose / limitation
NFC NDEF tag	MVP	Passive URL check-in. The secure dynamic NTAG 424 DNA is a production enhancement.
Temperature + humidity	MVP	Local heat/humidity trend and comfort indicator; protect from direct sunlight.
Light sensor	MVP	Measured brightness and sun/shade context.
LED bar / RGB LED	MVP	Physical display of local status and sensor fault condition.
UV sensor	Optional	Local UV-related indication after calibration; not a replacement for official UV alerts.
Noise level sensor	Optional	Relative sound level without recording or classifying conversations.
Barometer	Optional	Pressure trend; not a reliable weather forecast by itself.
Gas / air sensor	Research	Qualitative sensor response only unless an appropriate validated pollutant sensor is used.
NFC reader / PIR	Future	Card check-in or possible wake-up; PIR counts cannot reliably represent unique people.

5. Physical/link layer communication technologies

Short-range NFC: A passive ISO 14443 / NFC Forum tag is attached to each checkpoint. The visitor's phone supplies the NFC field; the tag needs no own battery. A standard NDEF URL launches a web address using the phone's operating-system capabilities on supported devices. We do **not** require the experimental Web NFC browser API, which has limited cross-browser support [7]. A visible QR code can be an accessibility fallback, although a static URL is easier to share remotely and requires the same security caveat.

Prototype uplink: The Raspberry Pi 5 uses campus Wi-Fi or a local hotspot (IEEE 802.11) to connect to the Python HTTP service. The Raspberry Pi 5 should use the USB-C power adapter. The MVP will run indoors or at an authorized supervised outdoor demonstration on mains power, not unsupervised on a battery.

Scalable uplink: A later terminal can replace the Raspberry Pi with a sleeping microcontroller, a LoRaWAN radio and a low-power sensor set. In the EU868 band the node could transmit compact environmental summaries every 10-15 minutes via a gateway. Gateway locations and radio coverage need a site survey, and airtime/duty-cycle constraints must be calculated. The Things Network also publishes fair-use limits; it should not be treated as an unlimited always-on bidirectional link [8]. Cellular NB-IoT or LTE-M is a fallback where available, with a higher recurring communications cost. NFC stamping can remain independent of the environmental network when the phone has Internet access.

Power and durability: The city-scale concept uses a photovoltaic panel, charger, rechargeable battery and a power budget confirmed by actual deep-sleep measurements. Sizing must account for winter irradiation, consecutive cloudy days, sensor warm-up, radio transmission and LED use. An IP65-class enclosure is a design target, with a ventilated/radiation-shielded compartment for measurements, UV-resistant labels, corrosion-resistant fixings and anti-vandal mounting. An enclosure must not block airflow or sunlight sensing. Installation near the Ría would require checks for flooding, maintenance access, permissions and protection of natural habitats.

6. Application/data layer communication technologies

The MVP uses **HTTPS + REST** with JSON, consistent with the course HTTP server. Example endpoints are shown below; these are our **proposed API**, not existing deployed services:

```
POST /api/checkins     {station_id, tag_payload, session_id}
GET  /api/progress     -> {month, stamps, reward_status}
POST /api/readings     {station_id, timestamp, temperature, ...}
GET  /api/stations     -> public conditions and route metadata
POST /api/redeem       {voucher_code, partner_id}
```

The backend authenticates devices and users separately, validates input, uses TLS and rejects repeated qualifying check-ins. A proposed monthly rule is: six qualifying stamps across at least two distinct stations, with no more than one qualifying stamp per station and day. A course demonstration with one physical node can simulate a second station identifier, actual physical presence at both places is not thereby proven.

In a future LoRaWAN deployment, payloads move from the radio gateway to a network server and then to our application using MQTT or an HTTP integration; LoRaWAN itself is not an HTTP transport. MQTT can also deliver occasional device configuration when nodes are reachable. Reward generation and partner redemption use HTTPS; a single-use code is marked redeemed atomically to avoid reuse.

7. Cloud-based monitoring dashboard and data analytics

NatureCheckpoint collects environmental measurements and records when users collect stamps at different stations. For the MVP, this data will be stored in a local SQLite database using Python. A simple web dashboard will allow us to monitor the stations, visualize sensor readings and track participation. We could also use Grafana to display environmental data and trends [9].

The final product will provide different information depending on who uses it:

Dashboard audience	Indicators
Visitors	Current environmental conditions at each checkpoint, available routes, collected stamps and progress towards rewards.
City and park operators	Number of check-ins per station, popular locations, environmental trends and technical status of the devices.
Business partners	Number of rewards issued and redeemed, allowing businesses to evaluate their participation.

By analyzing the collected data, NatureCheckpoint can identify which checkpoints are visited most often, how frequently users return and how environmental conditions influence participation. This information can help the city understand how the system is being used and which green spaces attract more interest. However, check-ins only represent NatureCheckpoint users, not the total number of visitors to a park.

The dashboard will also help us monitor the technical condition of each station. Missing sensor readings or connection problems can be detected early, making maintenance easier.

External data: We can combine our own measurements with publicly available environmental information. Bilbao's Anillo Verde dataset can help us plan routes and choose suitable checkpoint locations [1]. Air-quality information from Open Data Euskadi [10] and weather data from Euskalmet [11] could provide additional context. This would allow us to compare local measurements with regional conditions and give users more useful information before visiting a green space.

8. Ethics, privacy and environmental impact

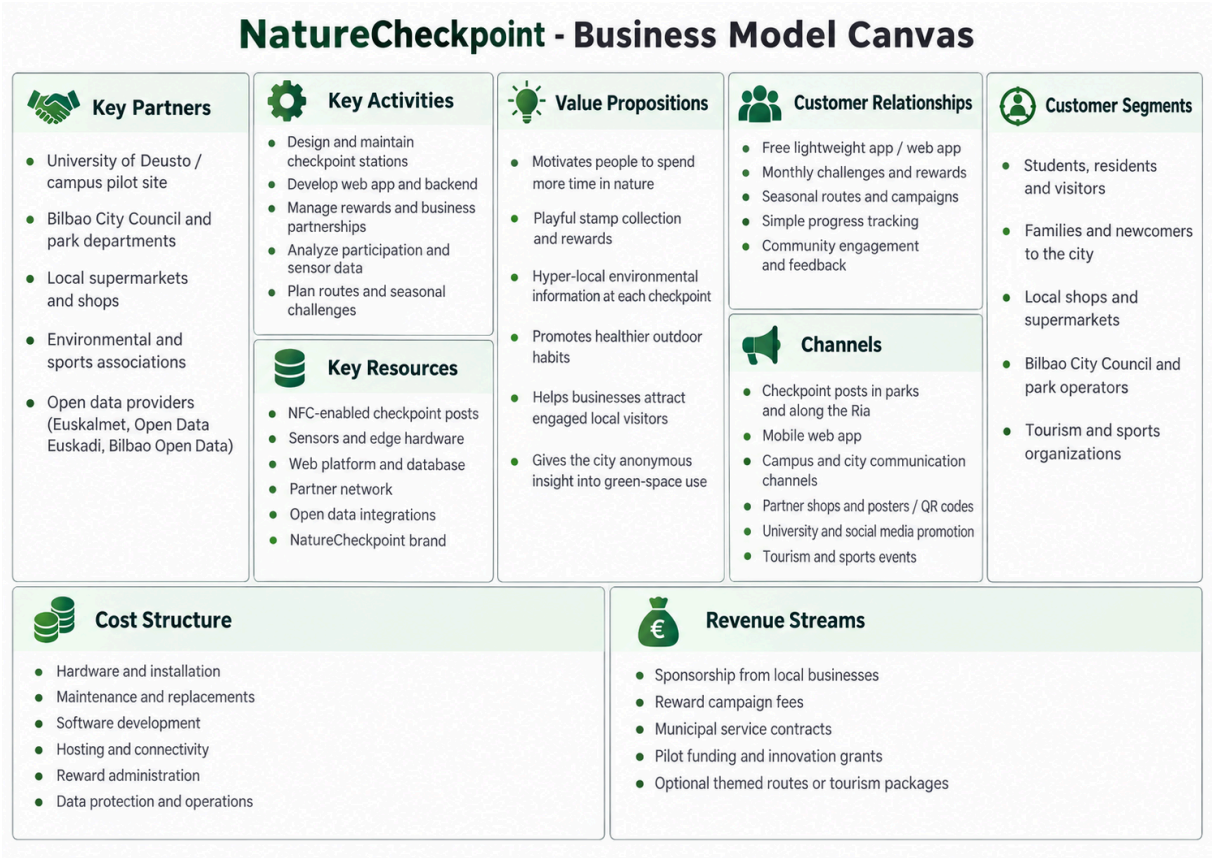
Privacy by design: The mobile web app does not request continuous GPS tracking. A visit nevertheless records a station and timestamp linked to a pseudonymous account, so it may constitute personal data; pseudonymization is not the same as anonymity. We propose no cameras, no audio recording and no behavioral profiling for advertisers. Public dashboards suppress small groups to reduce re-identification risk.

Safety and inclusion: Green-space access should never depend on owning a smartphone or buying from a partner. An NFC card/wristband or staffed-event route is a future inclusion option; the MVP would offer an alternative demonstration flow. Rewards must be modest and independent of speed or exertion. Route information will list known surfaces, steps and slopes only when validated. Severe weather, unsafe river conditions or a maintenance closure should override marketing prompts. LEDs communicate general information, not medical-grade advice or an assurance that air or UV is safe.

Environmental footprint: Solar energy does not eliminate the footprint of the panel, battery and electronics. A responsible design favors repairable modules, durable housings, low transmit frequency, minimal lighting, scheduled maintenance and proper battery/e-waste collection. Installations should be placed on existing infrastructure where possible, avoid sensitive habitats and be removed if the pilot ends. The app is deliberately brief: a check-in, a progress page and then attention returns to the landscape.

9. Business model

NatureCheckpoint uses a three-sided service model with a public institution or university as installation customer, residents as end users, and optional businesses as reward sponsors. **No partnerships, discounts or public funding are currently confirmed.**



The launch strategy is to validate engagement and technical reliability with a university pilot before offering a paid municipal service. Commercial viability depends on real installation and maintenance quotes, reward sponsorship commitments, the cost of preventing fraud and evidence of repeated use. Revenue and impact forecasts cannot be responsibly claimed without those tests. Sponsorship should not determine which routes are presented as environmentally better.

10. Initial task planning

We will work with short Scrum-style iterations, with a prioritized backlog, sprint reviews and a working increment after each stage. The group consists of Hischam, Florian and Julian. Detailed responsibilities can be assigned at the first planning meeting. The sequence below is a proposal, not a claim of completed work, and will be adjusted to the course schedule [12].

Sprint	Deliverable	Main activities / acceptance check
0	Proposal	Confirm problem and target group; obtain sensor availability and pilot permission; agree on reward rules.
1	Sensing	Connect Grove temperature/humidity and light sensors; log valid timestamped readings; LED reacts to them.
2	Stamping	Configure NFC URL; implement HTTPS check-in, pseudonymous session and duplicate protection; demonstrate a saved stamp.
3	Progress	Build mobile progress view and monthly threshold; simulate second station; issue a mock voucher.
4	Analytics	Create station dashboard, time-series view and aggregated check-in counts; integrate one verified open dataset if time permits.
5	Field test	Demonstrate complete workflow; observe users, test failure cases, document privacy and evaluate feasibility.

Definition of done for the MVP: (1) a sensor reading changes the local LED state; (2) an NFC phone tap opens the web app and a valid check-in persists in the backend; (3) duplicate attempts are handled according to the documented rules; (4) monthly progress unlocks a mock reward; (5) the dashboard displays sensor history and qualifying check-in counts; and (6) a visitor can finish the interaction and put the phone away.

Main risks and responses: If NFC phone behavior varies, test representative Android/iOS devices early and support a QR fallback. If a secure NTAG 424 DNA is not available, clearly label the standard-tag MVP as replayable. If a sensor lacks calibration or availability, reduce the inputs instead of fabricating an accurate comfort index. If an outdoor site cannot be approved, demonstrate the complete workflow in the lab with a supervised walk. Evaluate whether rewards actually encourage new outdoor activity, rather than merely attracting already-active visitors.

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